- **Policy as Code**—where policy is represented as machine-readable, version-controlled text using languages built to encode authorization policy.
- **Policy as Data**—where policy is represented and stored as structured data that represents relationships or other attributes needed to make an access decision.

The two approaches can be visualized as complementary ways of representing and evaluating policy as shown in figure 1.2.

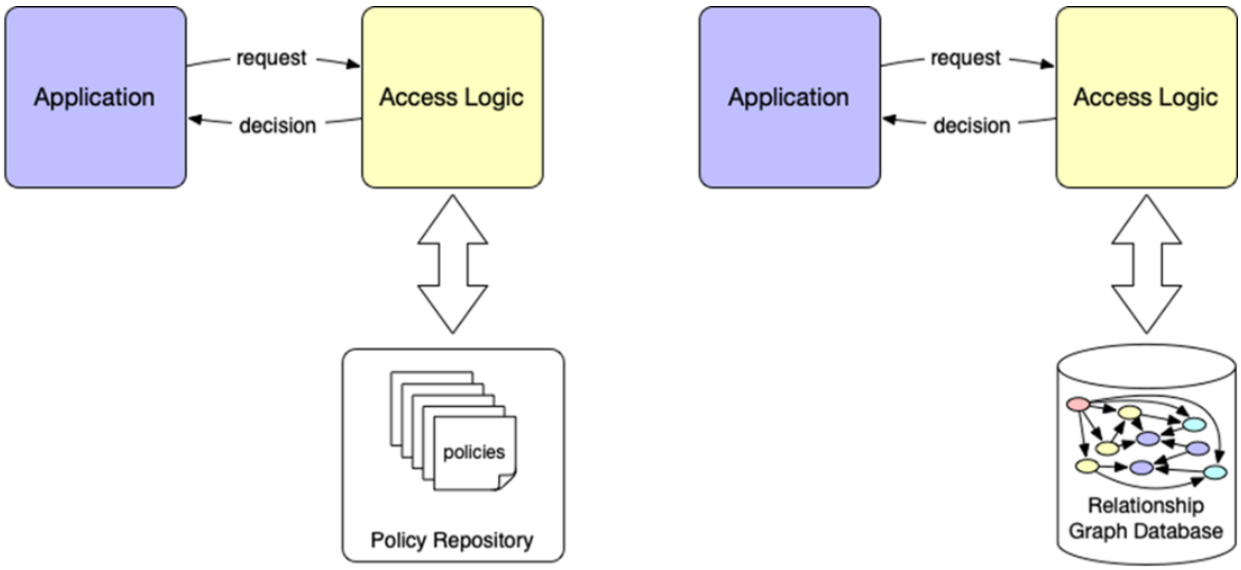


Figure 1.2 Dynamic authorization can represent policy in two complementary ways. On the left, Policy as Code stores machine-readable policies in a repository that the access logic evaluates at runtime. On the right, Policy as Data stores relationships and attributes in a structured data store that the same access logic uses to determine decisions. Both approaches externalize policy from the application while supporting different kinds of flexibility.

These two paradigms allow organizations to flexibly and efficiently manage access, but each addresses different needs.

1.4.1 Policy as Code: Defining access logic in a programmable way

Policy as Code treats access control rules as software. All the tools used for developing, testing, and deploying software can also be used to manage policies.